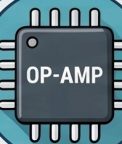


1. OP-AMP INTRODUCTION & PARAMETERS

INPUT MODES

- Differential
 - Sinif rejim
 - Common Mode
- each with input waveforms



ADVANCED PARAMETERS

- Input Impedance
- Output Impedance
- Voltage Gain
- Slew Rate

Impedances

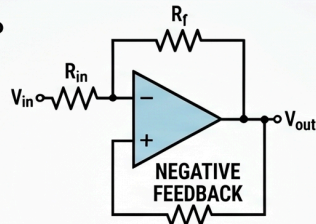
Input Bias Current

Input Offset Voltage

TASK 1: UNDERSTANDING OPERATIONAL AMPLIFIERS (OP-AMPS)

2. EFFECTS OF NEGATIVE FEEDBACK

OPEN-LOOP GAIN
(very high)



- Increases Stability
- Decreases Sensitivity
- Increases Linearity

NEGATIVE FEEDBACK LOOP

OPEN-LOOP GAIN
(very high)

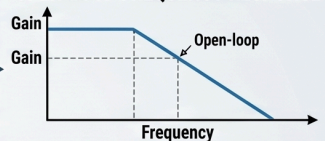
NEGATIVE FEEDBACK

STABLE, CONTROLLED GAIN

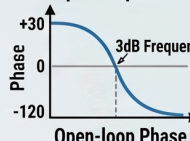
EXTENDED BANDWIDTH

3. FREQUENCY & PHASE RESPONSES

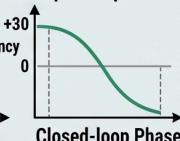
OPEN-LOOP FREQUENCY RESPONSE



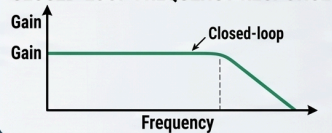
Open-loop Phase



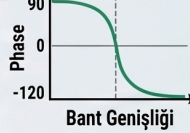
Open-loop Phase



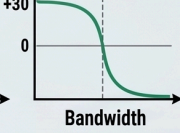
CLOSED-LOOP FREQUENCY RESPONSE



Open-loop Phase



Closed-loop Phase



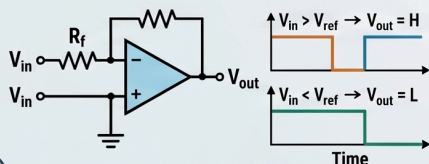
Bant Geniřlięi

Bandwidth

4. OP-AMP CONFIGURATIONS & APPLICATIONS

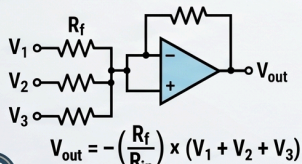
A

COMPARATOR



B

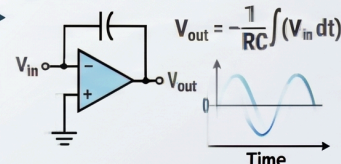
SUMMING AMPLIFIER



INTEGRATORS & DIFFERENTIATORS

C

INTEGRATOR



DIFFERENTIATOR

